JUNIOR SCHOOL LEARNING HUB

GRADE 8 MATHEMATICS — SCHEME OF WORK — TERM 2 2027

School: _____ Teacher: _____ TSC No.: _____

CBC planning format | Lesson outcomes are organised into Knowledge, Skills and Attitudes. Learning experiences retain the activities in the supplied Grade 8 scheme.

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
2	1	Measurements	Circles - Circumference of a circle	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define circumference as the distance around a circle **Skills:** Calculate the circumference using the formula C=πD or C=2πr **Attitudes:** Appreciate the relationship between diameter and circumference	**In groups, learners are guided to:** • Take a string and two sticks to draw circles on the ground • Measure the distance between fixed points • Use string and ruler to measure total length of line drawn • Compare diameter measurement with circumference	How do we determine the circumference of a circle?	- Master Mathematics Grade 8, pg. 81 - Strings - Sticks - Rulers - Circular objects	- Practical activities - Oral questions - Written exercises	
2	2	Measurements	Circles - Finding circumference of circular objects	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify circular objects in the environment **Skills:** Work out the circumference of different circular objects accurately **Attitudes:** Show interest in measuring circular objects	**In groups, learners are guided to:** • Discuss and find circumference of different circular objects in the environment • Complete tables to find missing measurements (radius, diameter, circumference) • Calculate circumference of bicycle wheels and clock hands • Solve real-life problems involving wheels and revolutions	Where do we find circles in our environment?	- Master Mathematics Grade 8, pg. 82 - Bicycle wheels - Clock models - Measuring tape - Circular objects	- Written tests - Practical work - Problem-solving	
2	3	Measurements	Circles - Length of an arc	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define an arc as a portion of circumference **Skills:** Calculate arc length using the formula Arc length = (θ/360) × 2πr **Attitudes:** Value the importance of arc calculations in real life	**In groups, learners are guided to:** • Make dummy clock using available resources • Trace path of minute hand in one revolution • Measure angles at centre and calculate arc lengths • Use cut outs to relate arcs to sectors	How do we calculate the length of an arc?	- Master Mathematics Grade 8, pg. 84 - Cartons for clock - Protractors - Strings - Rulers	- Practical exercises - Written assignments - Oral questions	
2	4	Measurements	Circles - Length of an arc	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define an arc as a portion of circumference **Skills:** Calculate arc length using the formula Arc length = (θ/360) × 2πr **Attitudes:** Value the importance of arc calculations in real life	**In groups, learners are guided to:** • Make dummy clock using available resources • Trace path of minute hand in one revolution • Measure angles at centre and calculate arc lengths • Use cut outs to relate arcs to sectors	How do we calculate the length of an arc?	- Master Mathematics Grade 8, pg. 84 - Cartons for clock - Protractors - Strings - Rulers	- Practical exercises - Written assignments - Oral questions	
2	5	Measurements	Circles - Perimeter of a sector	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain what a sector is and identify minor and major sectors **Skills:** Calculate perimeter of a sector using the formula: Perimeter = (θ/360 × 2πr) + 2r **Attitudes:** Show systematic approach in calculating sector perimeters	**In groups, learners are guided to:** • Draw circles and mark points to form sectors • Use string and ruler to determine arc length and add radii • Measure angles at centre • Calculate perimeter using formula and compare with measured values	How do we calculate the perimeter of a sector?	- Master Mathematics Grade 8, pg. 86 - Drawing instruments - Strings - Rulers - Protractors	- Written tests - Class activities - Problem-solving	
3	1	Measurements	Circles - Application and use of IT resources	**At the end of the lesson, the learner should be able to:** **Knowledge:** Discuss various applications of circles in real life **Skills:** Use IT or other resources to explore use of sectors and arcs **Attitudes:** Promote use of circles in real life situations	**In groups, learners are guided to:** • Solve problems involving merry-go-rounds, shot put areas • Calculate perimeters of semicircular objects • Use IT devices to explore circle applications • Work on complex problems involving multiple circles	How do we use circles in real life situations?	- Master Mathematics Grade 8, pg. 87 - Digital devices - Internet access - Real-life scenario cards	- Portfolio assessment - Presentations - Written assignments	
3	2	Measurements	Area - Area of a circle	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain how the formula for area of circle is derived **Skills:** Calculate area of a circle using the formula A = πr^2 **Attitudes:** Appreciate the importance of knowing circle areas	**In groups, learners are guided to:** • Draw and cut circles into equal sections • Arrange sections to form rectangle-like shape • Relate sides of rectangle to radius of circle • Work out area of rectangle formed	How do we calculate the area of a circle?	- Master Mathematics Grade 8, pg. 88 - Plain paper - Scissors - Rulers - Circular cut-outs	- Practical work - Written exercises - Oral questions	
3	3	Measurements	Area - Calculating areas of circles with different radii	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for area of a circle **Skills:** Calculate areas of circles given radius or diameter **Attitudes:** Show accuracy in area calculations	**In groups, learners are guided to:** • Calculate areas of circles with various radii • Find radius when area is given • Solve problems involving circular mats and grazing fields • Work out problems involving wire reshaping	What is the relationship between radius and area?	- Master Mathematics Grade 8, pg. 89 - Calculators - Worksheets - Problem cards	- Written tests - Problem-solving - Class activities	
3	4	Measurements	Area - Area of a sector of a circle	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define a sector as a fraction of a circle **Skills:** Calculate area of a sector using the formula: Area = (θ/360) × πr^2 **Attitudes:** Value precision in sector calculations	**In groups, learners are guided to:** • Draw circles and fold into equal parts • Calculate area using angle and radius • Use formula to find sector areas • Compare calculated areas with measured areas	How do we find the area of a sector?	- Master Mathematics Grade 8, pg. 91 - Drawing instruments - Protractors - Calculators - Paper for folding	- Written exercises - Practical activities - Oral questions	
3	5	Measurements	Area - Area of a sector of a circle	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define a sector as a fraction of a circle **Skills:** Calculate area of a sector using the formula: Area = (θ/360) × πr^2 **Attitudes:** Value precision in sector calculations	**In groups, learners are guided to:** • Draw circles and fold into equal parts • Calculate area using angle and radius • Use formula to find sector areas • Compare calculated areas with measured areas	How do we find the area of a sector?	- Master Mathematics Grade 8, pg. 91 - Drawing instruments - Protractors - Calculators - Paper for folding	- Written exercises - Practical activities - Oral questions	
4	1	Measurements	Area - Surface area of cubes	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain that a cube has 6 equal square faces **Skills:** Calculate total surface area using formula: TSA = 6 × length × length **Attitudes:** Show understanding of closed and open cubes	**In groups, learners are guided to:** • Study cubes and count number of faces • Measure sides of each face • Calculate area of each face • Derive formula for surface area of closed and open cubes	How do we calculate surface area of cubes?	- Master Mathematics Grade 8, pg. 92 - Cube models - Rulers - Measuring tape - Worksheets	- Written tests - Practical work - Problem-solving	
4	2	Measurements	Area - Surface area of cuboids	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify that cuboids have three pairs of equal rectangular faces **Skills:** Calculate surface area of cuboids systematically **Attitudes:** Appreciate applications of cuboid surface areas	**In groups, learners are guided to:** • Pick textbooks and measure length, width, height • Calculate area of each surface • Use models to understand pairs of equal sides • Derive formula for surface area	How is surface area of cuboid different from cube?	- Master Mathematics Grade 8, pg. 94 - Cuboid objects - Rulers - Cartons - Measuring instruments	- Written assignments - Class activities - Oral questions	

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4	3	Measurements	Area - Surface area of cylinders	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain that a cylinder opens to form two circles and a rectangle **Skills:** Calculate curved surface area using formula: CSA = 2πrh **Attitudes:** Show systematic approach in cylinder calculations	**In groups, learners are guided to:** • Select paper or plastic cylinders • Cut out top and bottom circles • Slit open hollow cylindrical part • Measure opened figure and relate to circumference	How do we find surface area of cylinders?	- Master Mathematics Grade 8, pg. 97 - Cylindrical objects - Scissors - Rulers - Paper cylinders	- Practical exercises - Written tests - Problem-solving	
4	4	Measurements	Area - Closed and open cylinders	**At the end of the lesson, the learner should be able to:** **Knowledge:** Distinguish between closed, open cylinders and pipes **Skills:** Calculate total surface area including circular ends **Attitudes:** Apply formulas to solve real-life problems	**In groups, learners are guided to:** • Calculate total surface area of closed cylinders • Work out surface area of open tanks and pipes • Solve problems involving petrol tanks and water pipes • Calculate surface area of semi-cylindrical troughs	When do we use different cylinder formulas?	- Master Mathematics Grade 8, pg. 99 - Cylinder models - Calculators - Real-life scenario cards	- Written assignments - Problem-solving - Class tests	
4	5	Measurements	Area - Closed and open cylinders	**At the end of the lesson, the learner should be able to:** **Knowledge:** Distinguish between closed, open cylinders and pipes **Skills:** Calculate total surface area including circular ends **Attitudes:** Apply formulas to solve real-life problems	**In groups, learners are guided to:** • Calculate total surface area of closed cylinders • Work out surface area of open tanks and pipes • Solve problems involving petrol tanks and water pipes • Calculate surface area of semi-cylindrical troughs	When do we use different cylinder formulas?	- Master Mathematics Grade 8, pg. 99 - Cylinder models - Calculators - Real-life scenario cards	- Written assignments - Problem-solving - Class tests	
5	1	Measurements	Area - Surface area of triangular prisms	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify the faces that make up a triangular prism **Skills:** Calculate surface area as sum of individual faces **Attitudes:** Value accuracy in prism calculations	**In groups, learners are guided to:** • Study triangular prism objects • Count number of faces • Identify triangular and rectangular faces • Calculate area of each face and find total	How do we calculate surface area of triangular prisms?	- Master Mathematics Grade 8, pg. 100 - Prism models - Rulers - Measuring instruments - Worksheets	- Written tests - Practical work - Oral questions	
5	2	Measurements	Area - Applications of triangular prisms	**At the end of the lesson, the learner should be able to:** **Knowledge:** Discuss real-life objects in the shape of triangular prisms **Skills:** Calculate surface areas of dust pans, tents, and goal posts **Attitudes:** Show interest in applying prism knowledge	**In groups, learners are guided to:** • Calculate surface area of rabbit hutches • Work out surface area of tents and dust pans • Solve problems involving wedges • Calculate surface area of handball goal posts covered with nets	Where do we find triangular prisms in real life?	- Master Mathematics Grade 8, pg. 102 - Real-life problem cards - Prism models - Calculators	- Written assignments - Problem-solving - Presentations	
5	3	Measurements	Area - Area of irregular shapes using square grids	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the method for estimating area of irregular shapes **Skills:** Estimate areas by counting full and partial squares **Attitudes:** Show patience in counting and estimating	**In groups, learners are guided to:** • Select graph paper and trace leaf outlines • Count number of full squares enclosed • Count partial squares and divide by 2 • Add full squares to half of partial squares	How do we estimate areas of irregular shapes?	- Master Mathematics Grade 8, pg. 103 - Graph paper - Square grids - Leaves - Pencils	- Practical activities - Written exercises - Observation	
5	4	Measurements	Area - Estimating areas of maps and other irregular shapes	**At the end of the lesson, the learner should be able to:** **Knowledge:** Apply square grid method to various irregular shapes **Skills:** Estimate areas of maps, assembly zones, and hand traces **Attitudes:** Promote use of area estimation in real life	**In groups, learners are guided to:** • Estimate area of fire assembly zones • Work out area of constituency maps • Estimate area of Kenya map • Trace palm of hand and estimate its area	What are practical uses of estimating irregular areas?	- Master Mathematics Grade 8, pg. 105 - Graph paper - Maps - Tracing paper - Calculators	- Portfolio assessment - Practical work - Written assignments	
5	5	Measurements	Money - Interest and principal	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define interest as extra money paid on borrowed amount **Skills:** Define principal as money borrowed **Attitudes:** Appreciate understanding of financial terms	**In groups, learners are guided to:** • Discuss amount of money that can be borrowed from mobile money providers • Calculate difference between amount borrowed and paid back • Identify institutions that offer loans • Complete tables relating principal, interest and total amount	What is interest in money?	- Master Mathematics Grade 8, pg. 107 - Sample loan documents - Calculators - Financial scenario cards	- Written exercises - Oral questions - Class activities	
6	1	Measurements	Money - Calculating simple interest	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain simple interest as money charged only on principal **Skills:** Calculate simple interest using formula: S.I = P × R × T / 100 **Attitudes:** Show accuracy in simple interest calculations	**In groups, learners are guided to:** • Discuss Mr. Murithi's loan scenario • Calculate total amount paid and interest • Express interest as percentage • Practice using formula with different values	How do we calculate simple interest?	- Master Mathematics Grade 8, pg. 109 - Calculators - Formula charts - Loan scenario cards	- Written tests - Problem-solving - Class activities	
6	2	Measurements	Money - Calculating simple interest	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain simple interest as money charged only on principal **Skills:** Calculate simple interest using formula: S.I = P × R × T / 100 **Attitudes:** Show accuracy in simple interest calculations	**In groups, learners are guided to:** • Discuss Mr. Murithi's loan scenario • Calculate total amount paid and interest • Express interest as percentage • Practice using formula with different values	How do we calculate simple interest?	- Master Mathematics Grade 8, pg. 109 - Calculators - Formula charts - Loan scenario cards	- Written tests - Problem-solving - Class activities	
6	3	Measurements	Money - Applications of simple interest	**At the end of the lesson, the learner should be able to:** **Knowledge:** Discuss various situations where simple interest applies **Skills:** Calculate amount paid back including interest **Attitudes:** Apply simple interest to solve real-life problems	**In groups, learners are guided to:** • Calculate interest for businessmen borrowing from financial institutions • Work out amount in bank accounts after interest • Find rate of simple interest from given information • Calculate interest earned on deposits	Where do we use simple interest in real life?	- Master Mathematics Grade 8, pg. 110 - Calculators - Real-life problem cards - Bank documents (samples)	- Written assignments - Problem-solving - Oral presentations	
6	4	Measurements	Money - Compound interest calculation step by step	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define compound interest as interest on principal and previous interest **Skills:** Calculate compound interest year by year up to three years **Attitudes:** Value systematic approach in compound interest	**In groups, learners are guided to:** • Discuss Mrs. Rono's investment in women groups • Calculate interest for first year and add to principal • Use new total as principal for second year • Continue process up to three years	How is compound interest different from simple interest?	- Master Mathematics Grade 8, pg. 112 - Calculators - Step-by-step charts - Comparison worksheets	- Written tests - Practical exercises - Class tests	
6	5	Measurements	Money - Working out appreciation per annum	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define appreciation as gain in value of a commodity **Skills:** Calculate appreciation using compound interest method **Attitudes:** Show understanding that appreciation is calculated like compound interest	**In groups, learners are guided to:** • Discuss meaning of appreciation in relation to monetary value • List items that appreciate in value • Calculate appreciation of land value year by year • Apply appreciation formula to various scenarios	What items appreciate in value and why?	- Master Mathematics Grade 8, pg. 115 - Calculators - Appreciation scenario cards - Charts	- Written exercises - Problem-solving - Oral questions	

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7	1	Measurements	Money - Working out depreciation per annum	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define depreciation as loss in value of a commodity **Skills:** Calculate depreciation step by step up to three years **Attitudes:** Appreciate that depreciation helps in making purchasing decisions	**In groups, learners are guided to:** • Discuss items that depreciate in value • Calculate depreciation of vehicles and electronics • Work through depreciation year by year • Compare depreciation with appreciation	What is depreciation and how do we calculate it?	- Master Mathematics Grade 8, pg. 116 - Calculators - Depreciation charts - Real-life examples	- Written tests - Class activities - Problem-solving	
7	2	Measurements	Money - Working out depreciation per annum	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define depreciation as loss in value of a commodity **Skills:** Calculate depreciation step by step up to three years **Attitudes:** Appreciate that depreciation helps in making purchasing decisions	**In groups, learners are guided to:** • Discuss items that depreciate in value • Calculate depreciation of vehicles and electronics • Work through depreciation year by year • Compare depreciation with appreciation	What is depreciation and how do we calculate it?	- Master Mathematics Grade 8, pg. 116 - Calculators - Depreciation charts - Real-life examples	- Written tests - Class activities - Problem-solving	
7	3	Measurements	Money - Hire purchase	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain hire purchase as buying goods through installments **Skills:** Calculate total cost under hire purchase **Attitudes:** Show consumer awareness in comparing cash and hire purchase prices	**In groups, learners are guided to:** • Visit places offering hire purchase or do online searches • Discuss different terms of purchase • Calculate installment periods and total amounts • Compare hire purchase prices with cash prices for consumer protection	How do we pay for goods on hire purchase?	- Master Mathematics Grade 8, pg. 117 - Hire purchase documents - Price comparison charts - Calculators	- Written assignments - Research projects - Oral presentations	
7	4	Measurements	Money - Visiting financial institutions and using IT for shopping	**At the end of the lesson, the learner should be able to:** **Knowledge:** Discuss information gathered from financial institutions **Skills:** Use IT to access online shopping platforms and identify terms of sale **Attitudes:** Spend money responsibly on needs and leisure	**In groups, learners are guided to:** • Visit or invite resource persons from banks and SACCOs • Gather information about interest rates offered on deposits • Use IT to access online shopping platforms • Discuss terms of sale for consumer awareness and protection	How do we make informed financial decisions?	- Master Mathematics Grade 8, pg. 118 - Digital devices - Internet access - Financial institution brochures - Guest speakers	- Portfolio assessment - Presentations - Reflection journals - Self-assessment	
7	5	4.0: Geometry	4.1: Geometrical Constructions - Constructing parallel lines using ruler and compasses 4.1: Geometrical Constructions - Constructing parallel lines using set square and ruler	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define parallel lines **Skills:** Construct parallel lines using a ruler and pair of compasses **Attitudes:** Appreciate the importance of accurate geometric constructions	**In groups, learners are guided to:** • Discuss the concept of parallel lines in real life • Follow step-by-step construction procedure using compass arcs • Draw a line and mark a point above it • Use compass arcs to construct parallel line through the point • Compare constructed lines with classmates	How can we construct parallel lines without measuring angles?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Pair of compasses - Pencil - Plain paper - Set square - Drawing paper	- Observation - Practical construction tasks - Oral questions	
8	1	4.0: Geometry	4.1: Geometrical Constructions - Constructing perpendicular bisector of a line 4.1: Geometrical Constructions - Constructing perpendicular from a point to a line using compasses	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define perpendicular bisector **Skills:** Construct perpendicular bisector using ruler and compasses **Attitudes:** Value accuracy in constructions	**In groups, learners are guided to:** • Draw a line of given length • Use compass to mark arcs from both ends • Identify intersection points of arcs • Join intersection points to form perpendicular bisector • Measure and verify equal segments and right angles	Why is the perpendicular bisector important in geometry?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Pair of compasses - Protractor - Pencil - Plain paper	- Observation - Practical construction - Written assignments	
8	2	4.0: Geometry	4.1: Geometrical Constructions - Constructing perpendicular using set square and ruler	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe the steps for constructing perpendiculars using set square **Skills:** Construct perpendicular lines using set square and ruler **Attitudes:** Show appreciation for geometric tools	**In groups, learners are guided to:** • Draw a horizontal line • Mark point above the line • Place ruler along the line • Position set square along ruler • Slide set square until edge touches the point • Draw perpendicular line through the point	What are practical applications of perpendicular lines in construction?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Set square - Ruler - Pencil - Drawing paper	- Observation - Practical construction - Peer review	
8	3	4.0: Geometry	4.1: Geometrical Constructions - Proportional division of a line 4.1: Geometrical Constructions - Sum of interior angles of polygons	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the method of dividing a line proportionally **Skills:** Apply the method of proportional division to divide lines into equal parts **Attitudes:** Demonstrate accuracy in geometric constructions	**In groups, learners are guided to:** • Draw line of given length • Draw auxiliary line at suitable angle • Mark equal intervals along auxiliary line using compasses • Join last point to end of original line • Draw parallel lines through other points • Verify equal divisions on original line	How can we divide a line without measuring its length?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Pair of compasses - Set square - Pencil - Protractor - Calculator	- Observation - Practical tasks - Written tests	
8	4	4.0: Geometry	4.1: Geometrical Constructions - Exterior angles of polygons 4.1: Geometrical Constructions - Constructing regular triangles	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define exterior angles of polygons **Skills:** Calculate sum of exterior angles and size of each exterior angle in regular polygons **Attitudes:** Appreciate the constant sum of exterior angles	**In groups, learners are guided to:** • Draw polygons and measure exterior angles • Calculate sum of exterior angles • Verify sum equals one complete revolution • Calculate exterior angle of regular polygons using formula • Complete table of polygon properties	Why is the sum of exterior angles always constant for any polygon?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Protractor - Ruler - Calculator - Chart showing polygon properties - Pair of compasses - Pencil	- Observation - Written tests - Problem-solving tasks	
8	5	4.0: Geometry	4.1: Geometrical Constructions - Constructing regular quadrilaterals (squares)	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe properties of squares **Skills:** Construct a square using ruler and compasses **Attitudes:** Demonstrate accuracy in perpendicular construction	**In groups, learners are guided to:** • Draw line of given length • Construct perpendicular at one end using compasses • Mark point along perpendicular • Use both ends as centers to locate fourth vertex • Join points to form square • Measure angles to verify right angles at each vertex	How do we ensure all angles in a square are right angles using only compass and ruler?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Pair of compasses - Protractor - Plain paper	- Observation - Practical tasks - Peer assessment	
9	1	4.0: Geometry	4.1: Geometrical Constructions - Constructing regular pentagons	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall that interior angle of regular pentagon is 108° **Skills:** Construct regular pentagon using ruler and protractor **Attitudes:** Show patience in multi-step constructions	**In groups, learners are guided to:** • Draw line of given length • Measure specified interior angle at one end • Mark point along the line at given distance • Repeat process at each new vertex • Join last vertex to starting point to complete pentagon • Verify all sides and angles are equal	Why is each interior angle of a regular pentagon 108°?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Protractor - Pencil - Calculator	- Observation - Practical construction - Written tests	
9	2	4.0: Geometry	4.1: Geometrical Constructions - Constructing regular pentagons	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall that interior angle of regular pentagon is 108° **Skills:** Construct regular pentagon using ruler and protractor **Attitudes:** Show patience in multi-step constructions	**In groups, learners are guided to:** • Draw line of given length • Measure specified interior angle at one end • Mark point along the line at given distance • Repeat process at each new vertex • Join last vertex to starting point to complete pentagon • Verify all sides and angles are equal	Why is each interior angle of a regular pentagon 108°?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Protractor - Pencil - Calculator	- Observation - Practical construction - Written tests	

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9	3	4.0: Geometry	4.1: Geometrical Constructions - Constructing regular hexagons and circles	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify that interior angle of regular hexagon is 120° **Skills:** Construct regular hexagon and circles related to triangles **Attitudes:** Appreciate relationship between circles and polygons	**In groups, learners are guided to:** • Construct regular hexagon using protractor • Construct triangle and draw perpendicular bisectors • Locate circumcenter and draw circumcircle • Construct angle bisectors to find incenter and draw incircle • Compare properties of different circles	How are circles related to regular polygons and triangles?	- MASTER Mathematics Grade 8 Learner's Book pg. 119 - Ruler - Protractor - Pair of compasses - Pencil	- Observation - Practical construction - Oral questions	
9	4	4.0: Geometry	4.2: Coordinates and Graphs - Drawing labelled Cartesian plane	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define Cartesian plane and identify its components **Skills:** Draw and label a Cartesian plane with axes and origin **Attitudes:** Show understanding of coordinate system	**In groups, learners are guided to:** • Draw horizontal line and label as x-axis • Draw vertical line crossing at center and label as y-axis • Mark intersection point as origin • Number axes with positive and negative values • Place arrows at ends of axes • Discuss purpose of arrows	Why do we need two axes to locate points on a plane?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Digital resources	- Observation - Oral questions - Written assignments	
9	5	4.0: Geometry	4.2: Coordinates and Graphs - Drawing Cartesian plane with different scales	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the concept of scale in graphs **Skills:** Draw Cartesian plane with specified scales on both axes **Attitudes:** Demonstrate accuracy in scaling	**In groups, learners are guided to:** • Draw Cartesian plane with various scales • Practice with different unit representations • Label axes correctly with chosen scale • Discuss when to use different scales • Compare graphs with different scales	How does scale affect the appearance of a graph?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Calculator	- Observation - Practical tasks - Written tests	
10	1	4.0: Geometry	4.2: Coordinates and Graphs - Identifying points on Cartesian plane	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe how to read coordinates of points **Skills:** Read coordinates of points on Cartesian plane correctly **Attitudes:** Show precision in reading coordinates	**In groups, learners are guided to:** • Draw Cartesian plane and mark points • Draw vertical line from point to x-axis to read x-coordinate • Draw horizontal line from point to y-axis to read y-coordinate • Write coordinates with x-value first, then y-value • Practice reading multiple points in different quadrants	How do we describe the exact position of a point on a plane?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Worksheet with points	- Observation - Oral questions - Written assignments	
10	2	4.0: Geometry	4.2: Coordinates and Graphs - Identifying points on Cartesian plane	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe how to read coordinates of points **Skills:** Read coordinates of points on Cartesian plane correctly **Attitudes:** Show precision in reading coordinates	**In groups, learners are guided to:** • Draw Cartesian plane and mark points • Draw vertical line from point to x-axis to read x-coordinate • Draw horizontal line from point to y-axis to read y-coordinate • Write coordinates with x-value first, then y-value • Practice reading multiple points in different quadrants	How do we describe the exact position of a point on a plane?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Worksheet with points	- Observation - Oral questions - Written assignments	
10	3	4.0: Geometry	4.2: Coordinates and Graphs - Plotting points on Cartesian plane	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process of plotting coordinates **Skills:** Plot given coordinates on Cartesian plane accurately **Attitudes:** Demonstrate accuracy in plotting	**In groups, learners are guided to:** • Identify x-coordinate and locate on x-axis • Check sign of y-coordinate • Draw line upward for positive y, downward for negative y • Locate y-coordinate on y-axis • Mark point where lines meet • Practice plotting points in all quadrants	How do we use coordinates to mark exact positions?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - List of coordinates	- Observation - Practical tasks - Peer assessment	
10	4	4.0: Geometry	4.2: Coordinates and Graphs - Reading coordinates from graphs	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify coordinates of plotted points on graphs **Skills:** Read coordinates from all four quadrants correctly **Attitudes:** Show accuracy in coordinate reading	**In groups, learners are guided to:** • Examine graph with plotted points • Write down coordinates of labeled points • Identify points on x-axis and y-axis • Match given coordinates to labeled points on graph • Practice with various coordinate positions	What special coordinates do points on the axes have?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper with plotted points - Ruler - Pencil - Practice worksheets	- Observation - Written tests - Oral questions	
10	5	4.0: Geometry	4.2: Coordinates and Graphs - Generating table of values from linear equations	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the process of generating tables from equations **Skills:** Generate table of values from given linear equations **Attitudes:** Show systematic approach to problem-solving	**In groups, learners are guided to:** • Choose suitable x values • Draw table with selected x values • Substitute each x value into equation to find y • Complete table with corresponding y values • Practice with equations in different forms	How do we find ordered pairs that satisfy a linear equation?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Pencil	- Observation - Written assignments - Problem-solving tasks	
11	1	4.0: Geometry	4.2: Coordinates and Graphs - Completing tables for linear equations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify given values in equation tables **Skills:** Complete given tables using equations accurately **Attitudes:** Demonstrate algebraic skills in context	**In groups, learners are guided to:** • Complete tables for equations in various forms • Substitute given values to find missing values • Generate complete tables for different equations • Practice with whole numbers and fractions • Verify completed tables	How do different forms of equations affect table generation?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Calculator - Pencil - Exercise book - Practice worksheets	- Observation - Written tests - Oral questions	
11	2	4.0: Geometry	4.2: Coordinates and Graphs - Completing tables for linear equations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify given values in equation tables **Skills:** Complete given tables using equations accurately **Attitudes:** Demonstrate algebraic skills in context	**In groups, learners are guided to:** • Complete tables for equations in various forms • Substitute given values to find missing values • Generate complete tables for different equations • Practice with whole numbers and fractions • Verify completed tables	How do different forms of equations affect table generation?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Calculator - Pencil - Exercise book - Practice worksheets	- Observation - Written tests - Oral questions	
11	3	4.0: Geometry	4.2: Coordinates and Graphs - Determining appropriate scale for graphs	**At the end of the lesson, the learner should be able to:** **Knowledge:** List factors to consider when choosing scales **Skills:** Choose suitable scales for given data ranges **Attitudes:** Show judgment in scale selection	**In groups, learners are guided to:** • Examine table with range of values • Consider graph paper size • Calculate range of values • Select scale that accommodates all values • Ensure efficient use of graph space	How do we choose a scale that makes best use of graph paper?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Data tables	- Observation - Practical tasks - Problem-solving	
11	4	4.0: Geometry	4.2: Coordinates and Graphs - Drawing line graphs from tables	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall steps for drawing line graphs **Skills:** Draw straight lines through plotted points using appropriate scale **Attitudes:** Show accuracy in graphing	**In groups, learners are guided to:** • Generate table of values using given equation • Choose suitable scale • Plot coordinates on Cartesian plane • Join plotted points using ruler • Draw line graphs for various equations • Verify line passes through all points	Why do linear equations produce straight line graphs?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Calculator	- Observation - Practical construction - Peer assessment	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
11	5	4.0: Geometry	4.2: Coordinates and Graphs - Drawing graphs for various linear equations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify equations representing horizontal and vertical lines **Skills:** Draw graphs for equations in different forms **Attitudes:** Demonstrate graphing skills	**In groups, learners are guided to:** • Draw graphs for equations in various forms • Draw horizontal and vertical lines • Compare slopes of different lines • Identify parallel and perpendicular lines • Practice graphing multiple equations	What do certain equations represent graphically?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Pencil - Set of equations	- Observation - Written tests - Practical tasks	
12	1	4.0: Geometry	4.2: Coordinates and Graphs - Introduction to simultaneous equations graphically	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define simultaneous equations **Skills:** Identify point of intersection of two lines as solution **Attitudes:** Show interest in graphical methods	**In groups, learners are guided to:** • Solve simultaneous equations algebraically • Draw graphs of both equations on same axes • Identify where lines intersect • Read coordinates of intersection point • Compare graphical solution to algebraic solution	How can graphs help us solve two equations together?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Number cards	- Observation - Oral questions - Written assignments	
12	2	4.0: Geometry	4.2: Coordinates and Graphs - Solving simultaneous linear equations graphically	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the graphical method for solving simultaneous equations **Skills:** Solve simultaneous equations using graphs accurately **Attitudes:** Demonstrate systematic approach	**In groups, learners are guided to:** • Generate tables for both equations • Choose appropriate scale for both equations • Plot both lines on same Cartesian plane • Identify point of intersection accurately • Write solution as ordered pair • Verify solution satisfies both equations	Why must the solution satisfy both equations?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Pencil	- Observation - Problem-solving - Written tests	
12	3	4.0: Geometry	4.2: Coordinates and Graphs - Solving simultaneous linear equations graphically	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the graphical method for solving simultaneous equations **Skills:** Solve simultaneous equations using graphs accurately **Attitudes:** Demonstrate systematic approach	**In groups, learners are guided to:** • Generate tables for both equations • Choose appropriate scale for both equations • Plot both lines on same Cartesian plane • Identify point of intersection accurately • Write solution as ordered pair • Verify solution satisfies both equations	Why must the solution satisfy both equations?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Pencil	- Observation - Problem-solving - Written tests	
12	4	4.0: Geometry	4.2: Coordinates and Graphs - Practice solving simultaneous equations with different forms	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify equations with decimal and fractional coefficients **Skills:** Solve various forms of simultaneous equations graphically **Attitudes:** Show proficiency in graphical methods	**In groups, learners are guided to:** • Solve equations with integer coefficients • Work with decimal coefficients • Handle equations with fractions • Practice with different forms • Compare solutions for accuracy	How do decimal coefficients affect the graphing process?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Scientific calculator - Pencil	- Observation - Written tests - Problem-solving tasks	
12	5	4.0: Geometry	4.2: Coordinates and Graphs - Applying simultaneous equations to real-life problems	**At the end of the lesson, the learner should be able to:** **Knowledge:** State real-life situations involving simultaneous equations **Skills:** Formulate and solve simultaneous equations from word problems graphically **Attitudes:** Appreciate practical applications of mathematics	**In groups, learners are guided to:** • Formulate equations from shopping scenarios • Set up equations from pricing problems • Solve using graphical method • Interpret solutions in context • Discuss other real-life applications	How do simultaneous equations help solve everyday problems?	- MASTER Mathematics Grade 8 Learner's Book pg. 147 - Graph paper - Ruler - Calculator - Real-life problem cards	- Observation - Problem-solving - Oral questions	